Compass Survey

Designation of Angles:

1. Sexagesimal System (India)

- 1 Circumference = 360°
- 1 degree = 60 minutes (60')
- 1 minute = 60 seconds (60")

2. Centesimal System (Europe)

- 1 Circumference = 400 grades
- 1 grade = 100 centigrade

3. Radian System

- 1 Circumference = 2π radians

4. Hour System

- 1 Circumference = 24 hours
- 1 hour = 60 minutes
- 1 minute = 60 seconds

Conversions:

- 1 circumference = 4 right angles = 360° = 400 grades = 2π radians = 24 hours
- 1 right angle = 90° = 100 grades = $\pi/2$ radians = 6 hours

Principles of Compass Survey:

- Based on traversing
- Fast
- Relatively Lower accuracy
- Suitable for large areas
- Captures more details (e.g., wooded areas)

Earth:

- Axis of rotation: North Pole / True North or Geographic North
- Earth has an imaginary bar magnet inside
- Magnetic field lines run from the Earth's magnetic South Pole to magnetic North Pole
- Magnetic needle points to Earth's magnetic North

Types of Meridians:

1. Magnetic meridian (used in compass survey) lines connecting magnetic north and magnetic south poles

2. True meridian (determined by astronomical observations, used by Survey of India for map preparation) lines connecting true north and true south poles

3. Arbitrary meridian (assumed direction connecting two permanent objects at the site, used for small area surveys)

Bearings:

Horizontal angle between a reference meridian and a survey line

1. Magnetic bearing (MB)
2. True bearing (TB)
3. Arbitrary bearing

Designation of Bearings:

1. Whole Circle Bearing (WCB)

- Clockwise angle from North

- Ranges from 0° to 360°

2. Quadrantal Bearing (QB) / Reduced Bearing (RB)

- Clockwise or anticlockwise angle measured from North or South towards East or West (whichever is nearest)

- Ranges from 0° to 90°

Q) WCB = 300°, Find QB = ?

Types of Compasses:

1. Prismatic Compass

2. Surveyor's Compass

Comparison of Compass Types:

Prismatic Compass	**Surveyor's Compass**
Broad type needle	Edge bar type needle
WCB from 0° to 360° (offset by 180°) North - 180° and South - 0°	QB from 0° to 90° (E and W interchanged) North - 0° and South - 0° East and West - 90°
More accurate due to simultaneous sighting and reading through prism	Less accurate, sighting done through sighting vane and reading through top
Needle and graduated ring attached and points North	Needle alone points North, graduations are on casing
Graduation are inverted	Graduation are erect
Tripod not necessary	Tripod essential
Does not act as index	Acts as index
Least count - 30"	Least count – 15"

→ Agate cap is fixed with prismatic compass

→ Prismatic compass is invented by captain kater.

Traverse:

- Network of instrument stations

- Fore bearing (FB): Direction of progression of survey

- Back bearing (BB): Direction opposite to progression of survey

In WCB,

$$BB = FB \pm 180°$$

$$FB = BB \pm 180°$$

Q) FB = 200, BB = ?

Q) BB = 10, FB = ?

In QB,

To find FB or BB just interchange direction.

FB of AB = S θ° E

BB of AB = N θ° W

Q) FB = N 50° W, find BB ?

Magnetic Declination

Magnetic declination is the horizontal angle between the magnetic meridian and true meridian at a given place. It varies from place to place and changes over time.

Cases of declination:

1. When magnetic north is west of true north: Declination is negative or westward
2. When magnetic north is east of true north: Declination is positive or eastward

Declination = 2°E or +2° (eastward)

3°W or -3° (westward)

Key terms:

- Isogonic lines: Lines joining points of equal declination
- Agonic lines: Lines joining points of zero declination

General formula: True Bearing (TB) = Magnetic Bearing (MB) ± declination

TB of sun at noon is always 0°, 180° or 360°.

Q) MB of sun at noon = 178°, find declination?

Variations in magnetic declination:

1. Secular Variation:

- Occurs over 250 years
- Magnitude: 5 to 10 minutes (oscillating like a pendulum)

2. Annual Variation:

- Occurs over 1 year
- Magnitude: 1 to 2 minutes
- Caused by Earth's revolution around the sun

3. Daily (Diurnal) Variation:

- Occurs daily
- Magnitude: Up to 10 minutes
- Caused by Earth's rotation
- It is maximum during day time than night, summer than winter and at poles than equator.

4. Irregular Variation:

- Magnitude: Up to 2 degrees.
- Caused by earthquakes, volcanic activity, magnetic storms

Magnetic Dip

Magnetic dip is the angle made by a freely suspended magnetic needle with the horizontal.

- At the North Pole: 90° dip downward
- At the Equator: 0° dip
- At the South Pole: 90° dip upward
- Northern Hemisphere: Dip downward
- Southern Hemisphere: Dip upward

Key terms:

- Isoclinic lines: Lines joining points of equal dip
- Aclinic line: Line joining points of zero dip (Equator is an example for aclinic line)

Local Attraction

Local attraction is the deflection of a magnetic needle from its actual position due to external magnetic forces.

Key points:

- All readings from a station affected by local attraction will have equal error.
- When a station is affected by local attraction:
  * All bearings will be incorrect
  * But included angles calculated will still be correct
- If Forward Bearing (FB) = Back Bearing (BB) ± 180°, both stations are free from local attraction
- If FB + BB ≠ 180°, at least one station is affected by local attraction

Causes of local attraction:

- Magnetic rocks, iron ore

- Electric lines, poles, railway lines
- Survey equipment (chains, arrows)